

Reprogramming Tumor Associated Macrophages Leads to T Cell and B Cell Interactions

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Abstract

Triple-negative breast cancer (TNBC) is an aggressive subtype of breast cancer defined by the absence of estrogen receptors, progesterone receptors, and HER2 receptor protein expression. The current standard of care combines immune checkpoint blockade with chemotherapy to activate anti-tumor immunity; however, stage TNBC still has a 77% 5-year survival rate. TNBC tumors often develop immunosuppressive microenvironments, characterized by a high abundance of tumor-associated macrophages (TAMs) and reduced infiltration of lymphoid cells, both of which contribute to poor patient prognosis. Previous work has shown that an MerTK inhibitor, MRX-2843, is a strong candidate for reprogramming TAMs to recruit increased T-Cell activation and infiltration in various cancers such as acute myeloid leukemia and glioblastoma. Currently, the Rosen lab is working to evaluate treatments integrating MRX-2843 for TNBC models. This combination treatment produced a durable long-term anti-tumor response (LTR), eliminating tumors in mice. When the same tumors were then re-introduced (mice were re-challenged), mice rejected the implanted tumor cells. This study aimed to identify the biological changes responsible for generating long-term immune memory following TNBC treatment. The response showed anti-tumor immunity due to increased infiltration of T cells, B cells, dendritic cells, and reprogrammed macrophages, increasing the amounts of cytotoxicity and antigen presentation in the tumor microenvironment. It was hypothesized that lymph nodes and the spleen would support sustained immune memory-based anti-tumor immunity by facilitating immune cell interactions. To test this, immunohistochemistry (IHC) was utilized to spatially resolve CD8 and CD4 T Cells, as well as B Cell localization within the TDLN and spleens of treated mice. Results demonstrate that combination-treated mice exhibit increased recruitment of CD4+T cells and CD20+ B cells within the spleen. In the

TDLNs, these mice further showed a marked reduction in F4/80+ macrophages compared to untreated tumor-bearing control mice. Additionally, further IHC revealed enhanced formation of germinal center-like structures composed of CD4+ T cells and CD20+ B cells. Notably, CD20+ B cells in LTR mice displayed migratory patterns consistent with movement from the follicular zone toward the T cell zone, indicating active crosstalk. These observations suggest that effective germinal center formation relies on coordinated interactions between CD4+ T cells and CD20+ B cells. Co-staining via immunofluorescence confirmed these interactions through the colocalization of CD4+ and CD20+ cells within germinal-center-like regions. Overall, MerTK regulated checkpoint therapy can be targeted to alter suppressive TNBC microenvironments to support adaptive immunity, increasing clinical efficacy. This information illustrates that the combination of MRX and CTX in treatment offers a strong anti-tumor immune response via facilitating TAM reprogramming to issue effective immunotherapeutic strategies to combat TNBC.

Introduction

Triple Negative Breast Cancer (TNBC) is defined by the absence of estrogen receptors, progesterone receptors, and HER2 receptor protein expression. As a result, it is a clinically aggressive and rapidly growing subtype of breast cancer that has a higher mortality rate than other cancers and accounts for 10-15% of breast cancer cases. TNBC is known to develop an immunosuppressive tumor microenvironment, utilizing different ways of evading the immune system. This can be seen by a high abundance of tumor-associated macrophages, (TAMs), which reduce infiltration of lymphoid cells, and contributes to poor patient prognosis. Specifically, TAMs lead to T Cell suppression, preventing them from activating, expanding, and infiltrating to attack the tumor cells. To properly treat TNBC with immunotherapy, the

current standard of care focuses on a combination therapy to reprogram TAMs to become recognized as antigen presenting cells by Cells, supporting T Cell activation towards tumor cell death. One target is the immune checkpoint molecule PD1, which is known to suppress host anti-tumor immunity in the tumor microenvironment. Currently, anti-PD1 antibodies have been shown to suppress PD1 and are used in clinical settings.

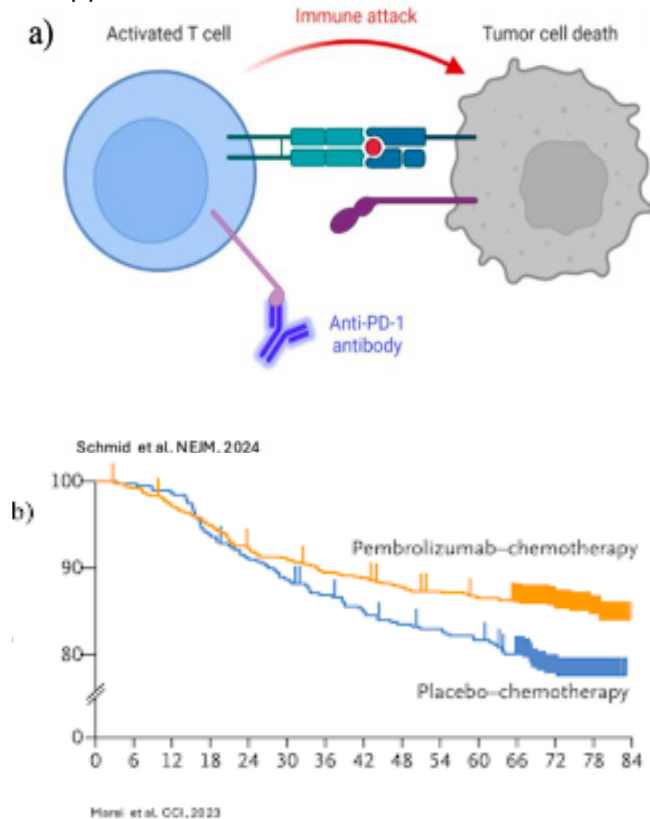


Figure 01. a) Visual representation of anti-PD-1 antibodies used to reprogram activated T-cells to target tumor cells. b) TNBC therapeutic standard of care uses anti-PD, showing increased survival rate.

An additional target for TAM reprogramming is MerTK receptors which are expressed on multiple immune cells, including TAMs. When MerTK is activated it suppresses inflammatory responses resulting in immunosuppression that allows tumors to evade immune detection and reduce T-cell activity. Previously, the Rosen lab at the Baylor College of Medicine Lester and Sue Smith Breast Center studied the effects of using MRX-2843 (MRX), a small-molecule MerTK inhibitor, in combination with the chemotherapy agent cyclophosphamide (CTX). It was shown that together, this treatment can re-educate the tumor microenvironment to increase T cell activation against cancer cells. This reprogramming of TAMs via MerTK can improve anti-PD1 therapy for TNBC patients.

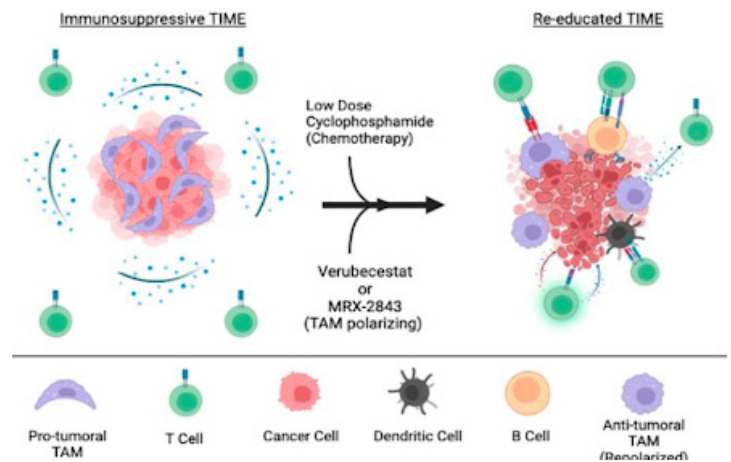


Figure 02. Low dose cyclophosphamide and MRX-2843 in combination treatment re-educates the tumor microenvironment from an immunosuppressive state to an immuno-active state.

When MRX was administered with CTX in 2153L p53-null TNBC tumor-bearing mice, this combination treatment produced a durable long-term anti-tumor response where mice were eliminated of cancer cells and resisted the rechallenged TNBC tumor cells. These triple combination treated mice exhibited a statistically significant higher survival rate than double treated mice without aPD1 and non-treated (Vehicle Control; VC) mice (Figure 03). Flow cytometry data from the tumors of combination treated vs control additionally revealed an increased number of reprogrammed TAMs and corresponding increased numbers of T Cells.

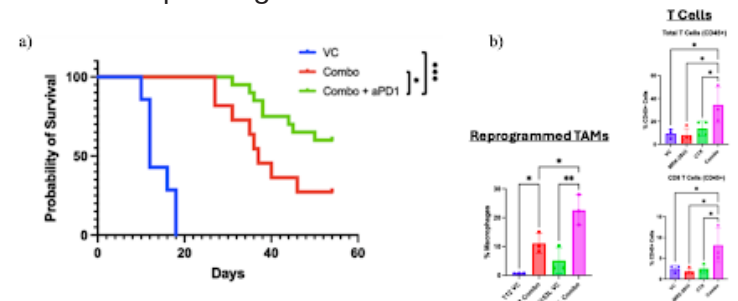


Figure 03. a) Survival plot demonstrating that triple combination treatment of MRX, CTX, and aPD1, generates a positive anti-tumor long-term response. b) Activated CD8+ and CD4 T Cells increased in 2153L COMBO treated tumors at Day 18.

The correlation between T cells in combination treated mice with increased survival conveyed a gap in the understanding of how the immune system was physically adapting; specifically, how the combination therapy led to systemic immune changes in lymphatic tissues.

Methods

Transplantable Syngeneic Mouse Models Mimic TNBC

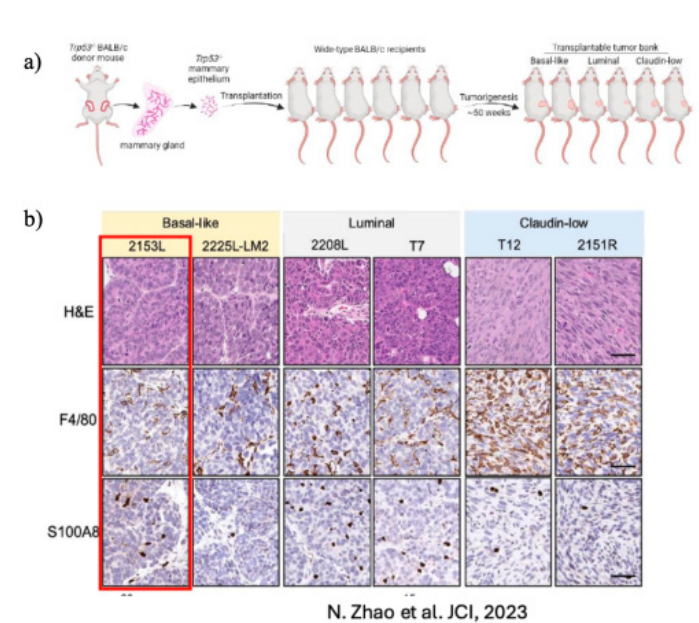


Figure 04. a) Representation of how syngeneic mouse models were made to mimic TNBC. b) H & E and IHC staining of different TNBC tumor lines.

This study utilized syngeneic mouse models which were developed using a p53-null BALB-C mouse. Knockout of p53 leads to lymphomas and osteosarcomas, but not breast cancers. In order to intentionally develop breast cancer, p53 null mammary cells were transplanted into the mammary fat pads of dozens of wildtype BALB-C mice which grew tumors that were harvested to create a tumor cell line bank that included a variety of different tumors (Basal-like, luminal, claudin-low, etc). For this study, the 2153L tumor line was used.

Spleen Analysis

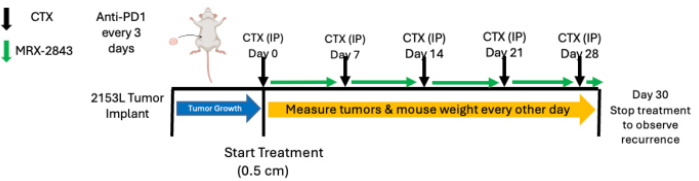


Figure 05. Timeline of combination treated mouse study to analyze effects on the spleen.

2153L tumors were implanted in p53-null mice and allowed to grow for one week before starting combination treatment (CTX, MRX, and anti-PD1). On day 18, spleens were harvested from Vehicle Control (non-treated; abbr. VC) and combination treated (COMBO) mice. Sections were sliced using a microtome and then stained with IHC with validated rabbit and rat antibodies.

Lymph Node Analysis

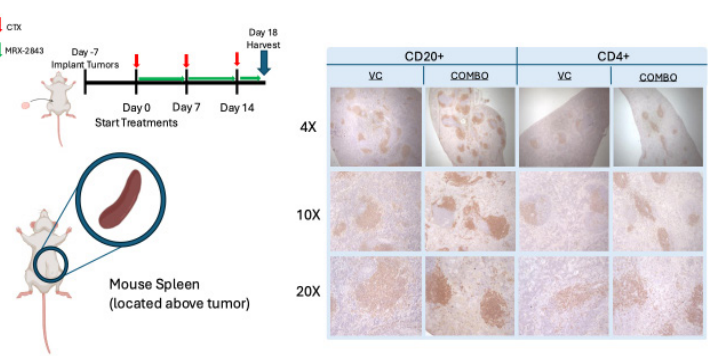


Figure 06. Timeline of combination treated mouse study to analyze effects on the tumor draining lymph node.

To evaluate long-term immune responses, Tumor Draining Lymph Nodes (TDLN) from long-term responder (triple combination treatment of CTX, MRX, and anti-PD1 and survived, LTR) mice were harvested. LTR mice showed the highest survival rate after triple combination treatment (Figure 03). TDLN sections were sliced with a microtome and stained with IHC and immunofluorescence for macrophages, CD8⁺ cytotoxic T cells, CD4⁺ T cells, and CD20⁺ B cells. B220 was also validated as an alternative B cell marker for co-staining with CD4. Imaging focused on identifying cell localization and potential germinal center-like structures.

Results and Discussion

Immunohistochemistry (IHC) from the 2153L spleens showed remodeling of the lymph tissue. The combination-treated mice displayed an increase of T cells compared to the vehicle control (non-treated) mice (Figure 07). Furthermore, CD4⁺ T cells and CD20⁺ B cells in the triple-combination (COMBO) treated spleens visually appeared to be more clustered in a germinal-center like pattern compared to the weaker clustering shown in the vehicle control (Figure 07). This splenic remodelling suggested systemic changes in lymphatic tissue, shifting focus towards how tumor-draining lymph nodes (TDLNs) sustain memory-driven anti-tumor responses through coordinated cell interactions.

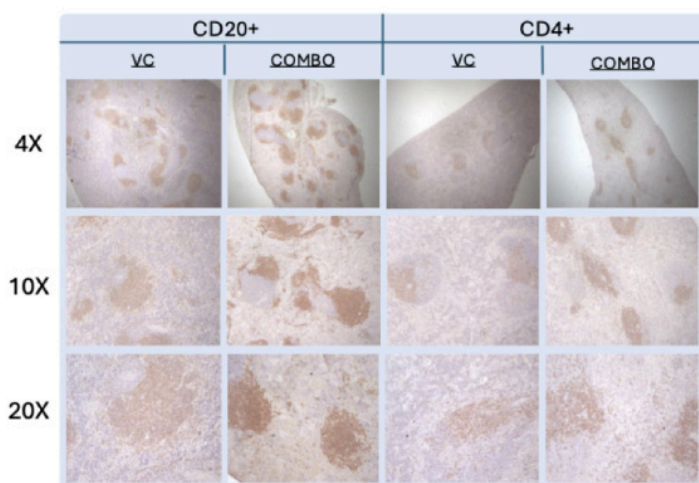


Figure 07. Remodeled spleen tissue from Day 18 in combination treated versus untreated vehicle control 2153L mice.

IHC of sections of the TDLN suggested a decrease of macrophages and an increase of CD8+ cytotoxic T Cells in the LTR mice compared to NTB mice. Additionally, further IHC revealed enhanced formation of germinal center-like structures composed of CD4+ T cells and CD20+ B cells. Notably, CD20+ B cells in LTR mice displayed migratory patterns consistent with movement from the follicular zone on the outside of the lymph, towards the T cell zone, indicating active crosstalk between the B cells and T cells. These observations suggest that effective anti-tumor long term response may rely on coordinated interactions between CD4+ T cells and CD20+ B cells to initiate memory. In order to validate this claim, a visualization of B Cells and T Cells together was necessary for identification of colocalization (whether or not B and T Cells were grouped together).

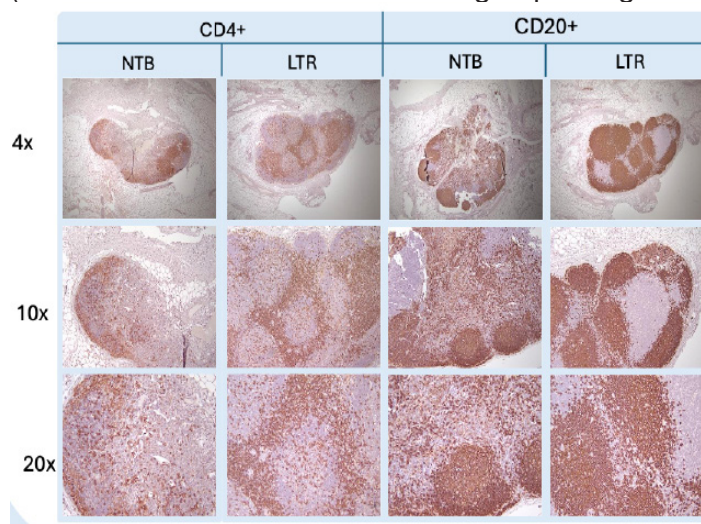


Figure 08. CD4+ T Cells and CD20+ B cells show more clustering in long-term responding (LTR) mice compared to non-tumor bearing (NTB) 2153L mice.

Utilizing immunofluorescence yielded a visual representation of the colocalized clusters of T-Cells and B-Cells in the LTR mice (Figure 09). In LTR mice, these clusters appeared to migrate inwards towards the T-Cell zone of the lymph node, whereas in the NTB mice, these co-localized clusters appear to predominantly remain on the follicular zone towards the outer edge of the lymph node (Figure 09a). To better validate these results, IHC was run with a separate group of rat antibodies and validated the B220 antibody for IF. Once validated, B220 and CD4 were used to co-stain for B cells and T Cells. The LTR clusters show both migratory patterns and increased co-localization of CD4 tangled with the B Cells in the cluster compared to the NTB (Figure 09b).

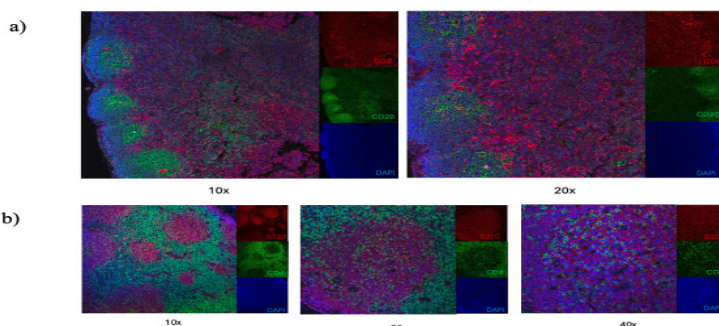


Figure 09. a) Non-Tumor Bearing (NTB) mice show B Cell Clusters predominantly in the follicular zone of the TDLN. CD20 displayed in green, CD8 displayed in red. b) LTR mice exhibit B220 B cells and CD4 T cells clustering in close proximity migrating to the T cell zone. CD4 displayed in green, B220 displayed in red.

This observed migration of CD20+ B cells from the follicular zone into the T-cell zone of the TDLN suggests the active interactions between B cells and CD4+ T helper cells, leading to potential formation of germinal center-like structures. This spatial reorganization is indicative of B cell activation and antigen presentation, a process critical for the development of high-affinity antibody responses and memory T cell formation. In the context of MRX, CTX, and aPD1 triple-combo treatment, this migration supports a model in which these specific interactions with B cells function as antigen-presenting cells (APCs) that help sustain a long-term, memory-based anti-tumor immune response. This finding reinforces the role of adaptive immune coordination facilitated by TAM reprogramming in order to achieve durable immunotherapy outcomes in TNBC.

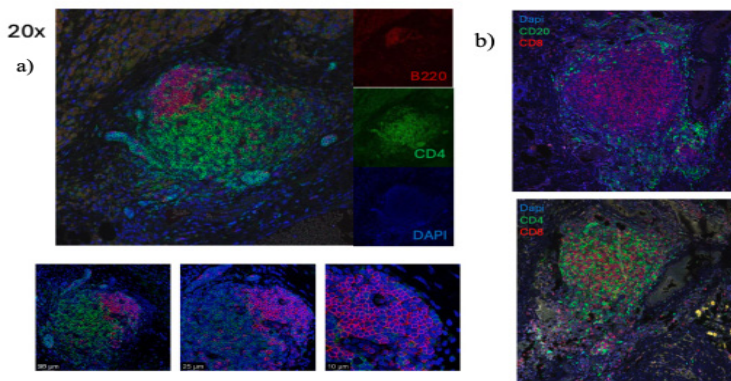


Figure 10. a) Formation of a tertiary lymphoid structure composed of B220 and Cd4 cells found in the residual non-lymphatic tissue of the tumor bed in LTR mice. b) A second TLS previously found by the Rosen lab in separate LTR mouse images with CD8+ cytotoxic T-Cells, CD20+ B-Cells, and CD4+ helper T cells.

Another key finding of this study regarded the formation of lymphoid aggregates. Previous studies have shown that tertiary lymphoid structures (TLS) formation is a positive biomarker of successful TNBC treatment with anti-PD19. They have been found in tissues of patients who have responded well to TNBC immunotreatment, whereas patients who did not respond well, do not exhibit this TLS formation¹⁰. While imaging the LTR mice, a lymphoid aggregate, thought to be a TLS, composed of B220+ cells and CD4+ cells was found in the residual tumor bed of a LTR mouse (Figure 10a). This TLS may be a strong plausible source of the LTRs active immune system targeting TNBC, forming the immune memory this treatment aims to generate.

Conclusions

These results are promising; however, biological replicates will be needed to further validate the TDLN results for the LTRs. Additionally, equal replications with the appropriate Non-Tumor Bearing controls will be necessary. Moreover, to analyze the data with more efficiency and accuracy, future actions should be taken to scan the spleen IHC slides and quantify the number of marked cells such as CD20+ B cells and CD4+ T cells with a machine learning program. The TDLN and co-localized clusters should be measured not only in quantity, but also in their respective distance from the center of the TDLN to the edge of it to quantify the observed migration patterns. Concurrently, the lymphoid aggregate structure the mechanisms by which TLSs form remain poorly understood, though promising. Due to the nature of their small size and the way tissue samples are collected and sampled, it is difficult to find these lymphoid aggregates and TLS structures and discern their difference. Further staining of the TDLN should be conducted to

identify more of these formations, and better methods in doing so should be generated. Understanding the formation and role of these TLS will aid with developing a better understanding if B cells are actually the key acting antigen-presenting cells in the tumor microenvironment, as well as respectively identifying the specific subsets of B-Cells are critical to activate and support T cells in anti-tumor immunity, as well as identifying if these clusters are in fact germinal centers.

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References

1. Cortes, Jose, Hope S. Rugo, Diana W. Cescon, Sun A. Im, Mehwish M. Yusof, Cristina Gallardo, et al. "Pembrolizumab plus Chemotherapy in Advanced Triple-Negative Breast Cancer." *New England Journal of Medicine* 387, no. 3 (2022): 217–26.
2. Rey-Vargas, L., M. C. Sanabria-Salas, L. Fejerman, and S. J. Serrano-Gomez. "Risk Factors for Triple-Negative Breast Cancer among Latina Women." *Cancer Epidemiology, Biomarkers & Prevention* 28, no. 11 (2019): 1771–83.
3. Singh, S., N. Lee, D. A. Pedroza, I. L. Bado, C. Hamor, L. Zhang, S. Aguirre, et al. "Chemotherapy Coupled to Macrophage Inhibition Induces T-cell and B-cell Infiltration and Durable Regression in Triple-Negative Breast Cancer." *Cancer Research* 82, no. 12 (June 15, 2022): 2281–2297. <https://doi.org/10.1158/0008-5472.CAN-21-3714>
4. Lee-Sherick, A. B., K. M. Jacobsen, C. J. Henry, M. G. Huey, R. E. Parker, L. S. Page, et al. "MERTK Inhibition Alters the PD-1 Axis and Promotes Anti-Leukemia Immunity." *JCI Insight* 5, no. 23 (2020).
5. Su, Y. T., M. Butler, M. Zhang, W. Zhang, H. Song, L. Hwang, et al. "MerTK Inhibition Decreases Immune Suppressive Glioblastoma-Associated Macrophages and Neoangiogenesis

- in Glioblastoma Microenvironment.” *Neuro-Oncology Advances* 2, no. 1 (2020): vdaa065.
6. NBCF Team. “Triple Negative Breast Cancer.” National Breast Cancer Foundation. January 15, 2026. <https://www.nationalbreastcancer.org/triple-negative-breast-cancer/>.
 7. Liu, Z., M. Li, Z. Jiang, and X. Wang. “A Comprehensive Immunologic Portrait of Triple-Negative Breast Cancer.” *Translational Oncology* 11, no. 2 (April 2018): 311–329. <https://doi.org/10.1016/j.tranon.2018.01.011>.
 8. Zhao, N., E. B. Kabotyanski, A. B. Saltzman, A. Malovannaya, X. Yuan, L. C. Reineke, N. Lieu, et al. “Targeting eIF4A Triggers an Interferon Response to Synergize with Chemotherapy and Suppress Triple-Negative Breast Cancer.” *Journal of Clinical Investigation* 133, no. 24 (December 15, 2023): e172503. <https://doi.org/10.1172/JCI172503>.
 9. Vanhersecke L, Brunet M, Guégan JP, Rey C, Bougouin A, Cousin S, Moulec SL, Besse B, Lorient Y, Larroquette M, Soubeyran I, Toulmonde M, Roubaud G, Pernot S, Cabart M, Chomy F, Lefevre C, Bourcier K, Kind M, Giglioli I, Sautès-Fridman C, Velasco V, Courgeon F, Oflazoglu E, Savina A, Marabelle A, Soria JC, Bellera C, Sofeu C, Bessede A, Fridman WH, Loarer FL, Italiano A. Mature tertiary lymphoid structures predict immune checkpoint inhibitor efficacy in solid tumors independently of PD-L1 expression. *Nat Cancer*. 2021 Aug;2(8):794-802. doi: 10.1038/s43018-021-00232-6. Epub 2021 Aug 12. PMID: 35118423; PMCID: PMC8809887.
 10. Schumacher TN, Thommen DS. Tertiary lymphoid structures in cancer. *Science*. 2022 Jan 7;375(6576):eabf9419. doi: 10.1126/science.abf9419. Epub 2022 Jan 7. PMID: 34990248.